

Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

Class Level Information		
Class	Levels	Values
PRESSURE	2	25 30
SPEED	2	200 250
CARBONATION	3	10 12 14
REPLICATE	2	1 2

Number of Observations Read	24
Number of Observations Used	24

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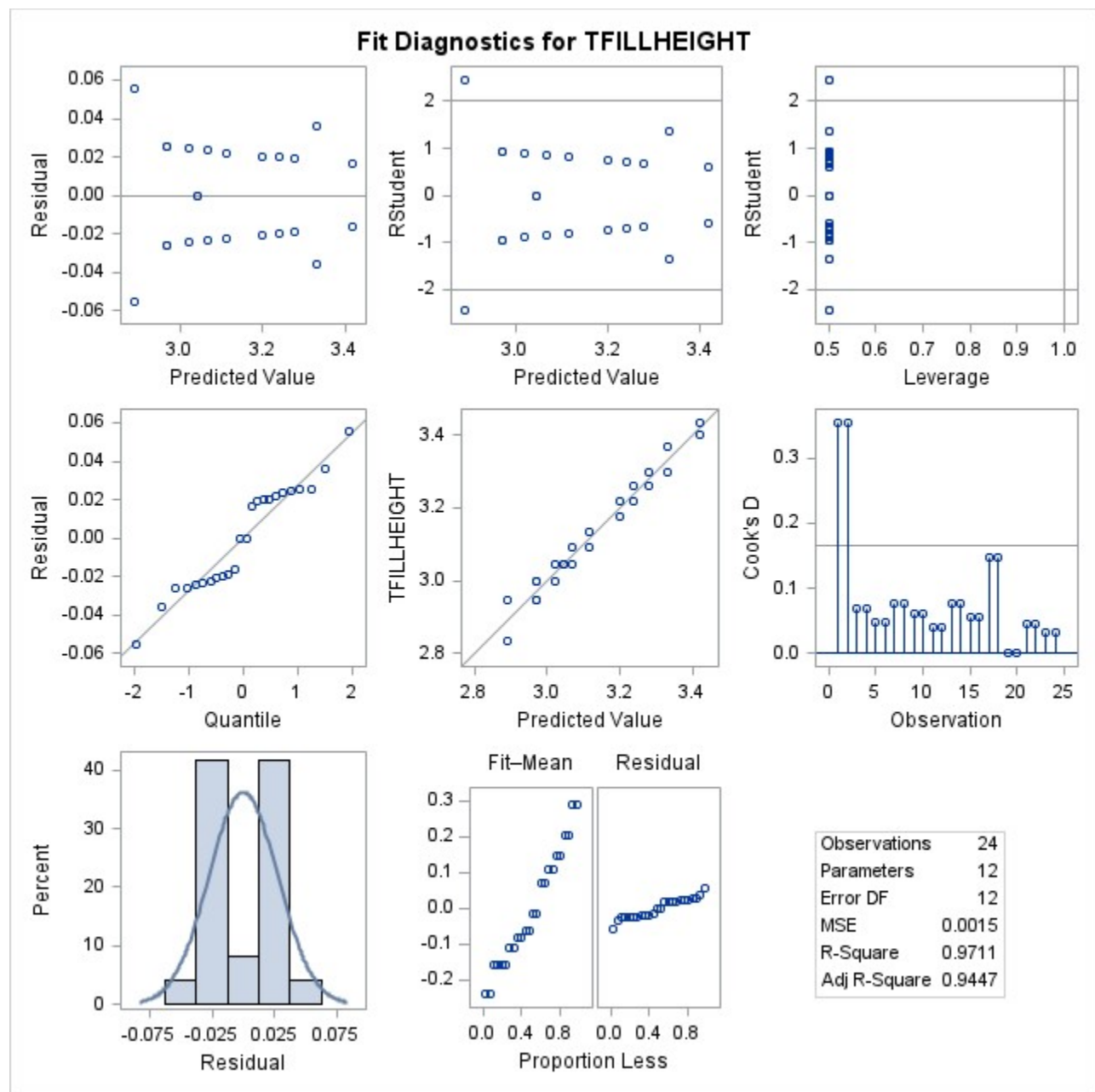
Dependent Variable: TFILLHEIGHT

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	0.58838494	0.05348954	36.72	<.0001
Error	12	0.01748214	0.00145685		
Corrected Total	23	0.60586708			

R-Square	Coeff Var	Root MSE	TFILLHEIGHT Mean
0.971145	1.220167	0.038169	3.128148

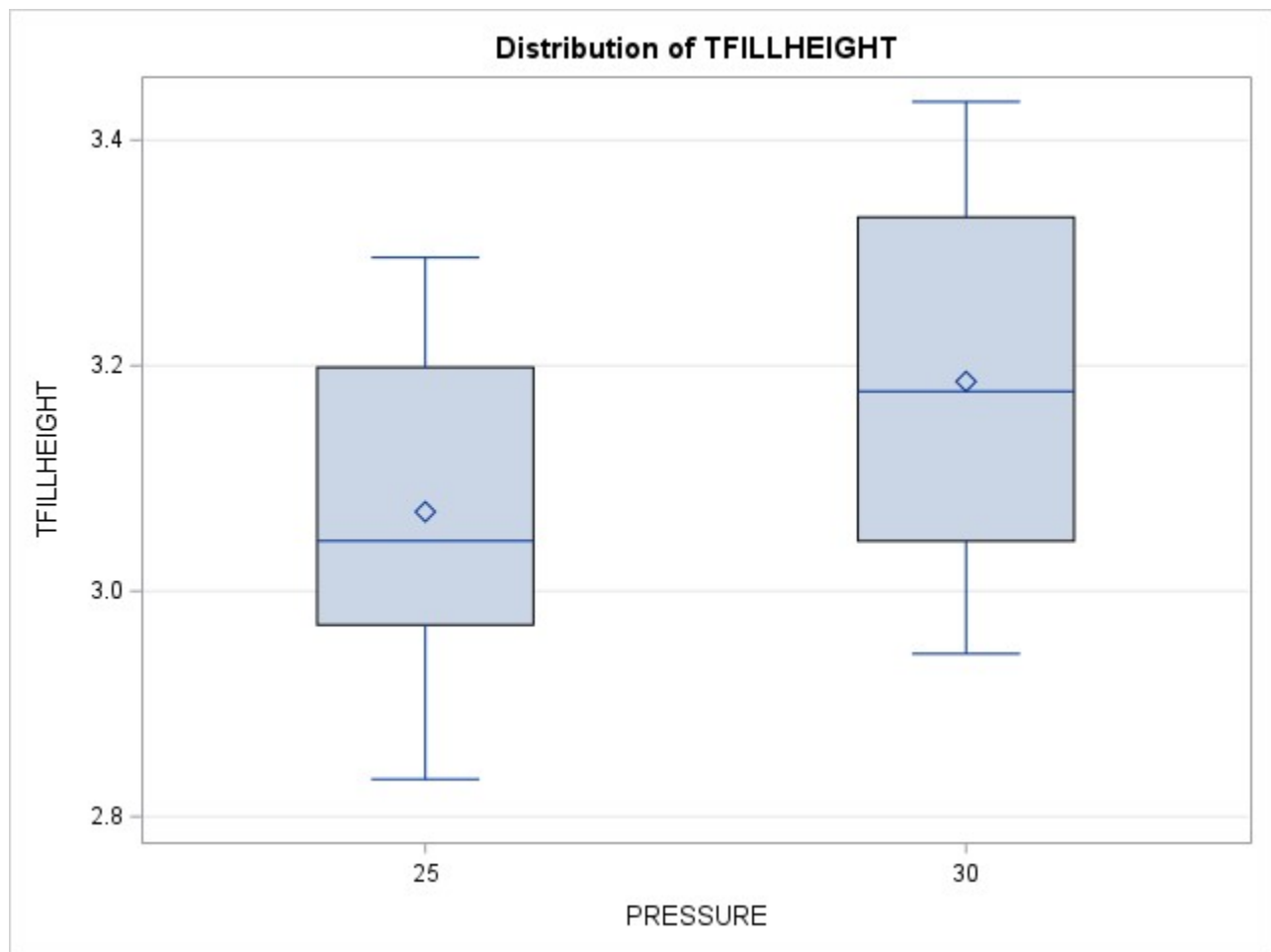
Source	DF	Type I SS	Mean Square	F Value	Pr > F
PRESSURE	1	0.08010344	0.08010344	54.98	<.0001
SPEED	1	0.04052411	0.04052411	27.82	0.0002
PRESSURE*SPEED	1	0.00102088	0.00102088	0.70	0.4189
CARBONATION	2	0.46033653	0.23016826	157.99	<.0001
PRESSURE*CARBONATION	2	0.00428750	0.00214375	1.47	0.2682
SPEED*CARBONATION	2	0.00007378	0.00003689	0.03	0.9750
PRESSU*SPEED*CARBONA	2	0.00203869	0.00101935	0.70	0.5159

Source	DF	Type III SS	Mean Square	F Value	Pr > F
PRESSURE	1	0.08010344	0.08010344	54.98	<.0001
SPEED	1	0.04052411	0.04052411	27.82	0.0002
PRESSURE*SPEED	1	0.00102088	0.00102088	0.70	0.4189
CARBONATION	2	0.46033653	0.23016826	157.99	<.0001
PRESSURE*CARBONATION	2	0.00428750	0.00214375	1.47	0.2682
SPEED*CARBONATION	2	0.00007378	0.00003689	0.03	0.9750
PRESSU*SPEED*CARBONA	2	0.00203869	0.00101935	0.70	0.5159



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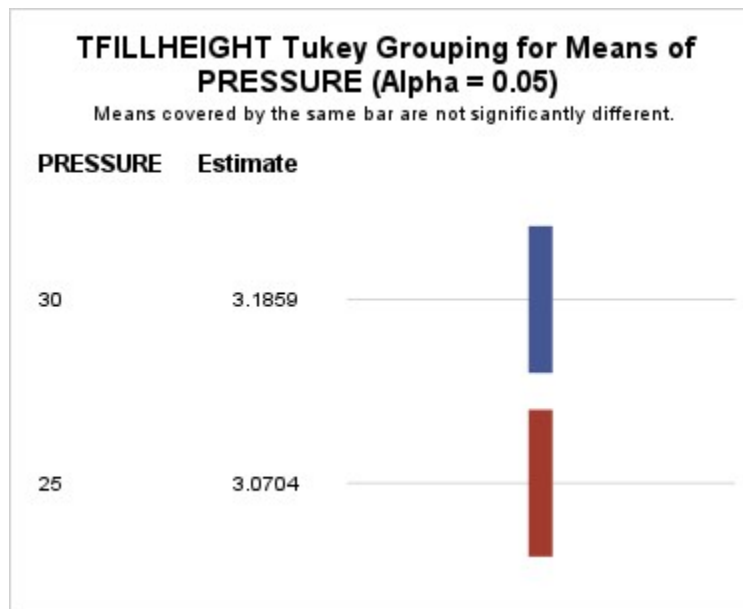
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The GLM Procedure

Tukey's Studentized Range (HSD) Test for TFILLHEIGHT

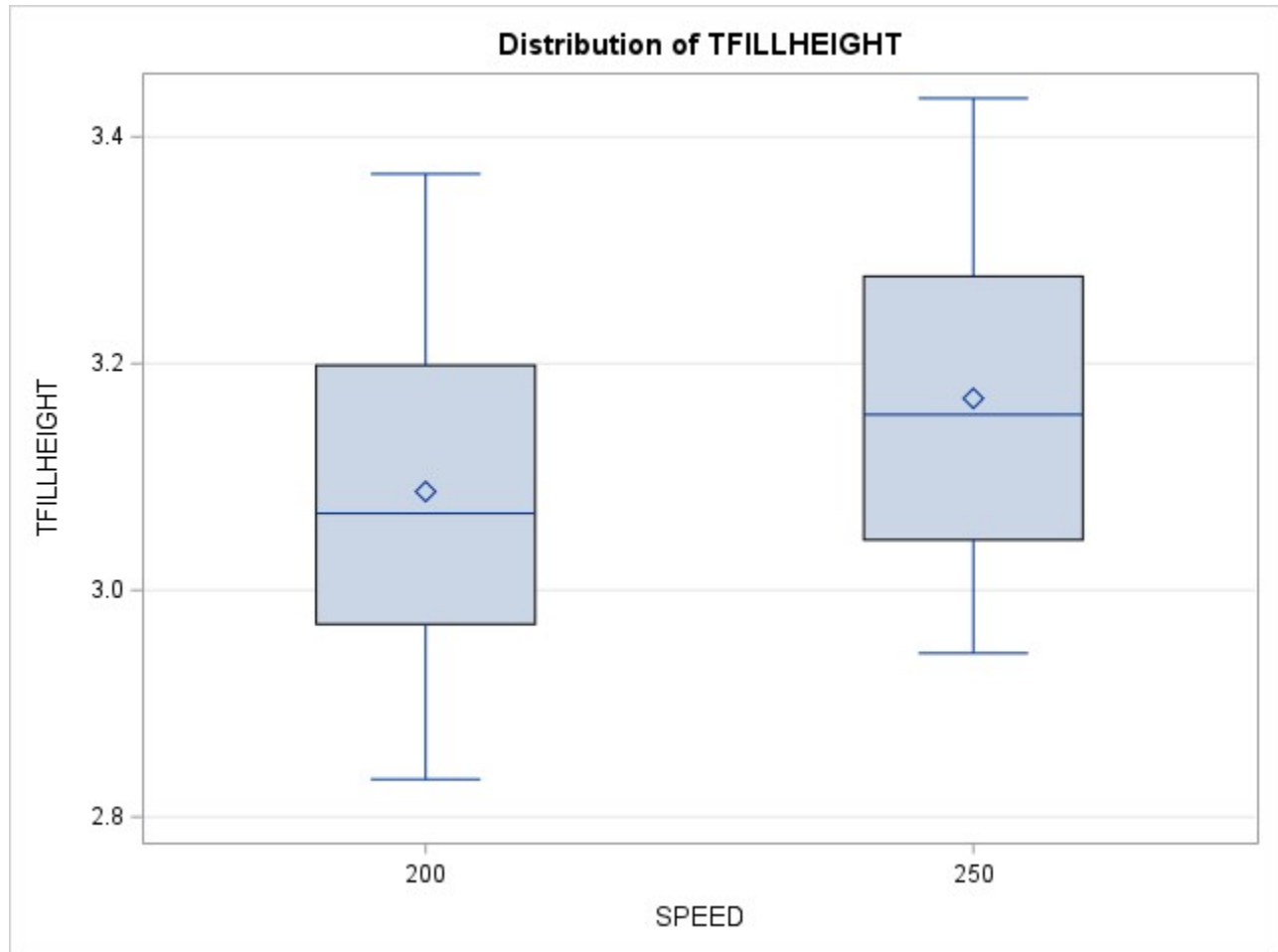
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.001457
Critical Value of Studentized Range	3.08118
Minimum Significant Difference	0.0339



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The GLM Procedure



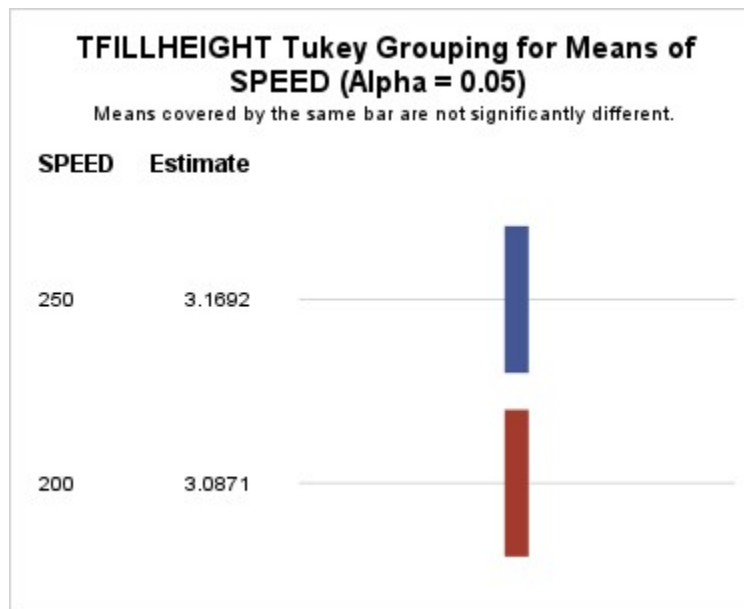
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The GLM Procedure

Tukey's Studentized Range (HSD) Test for TFILLHEIGHT

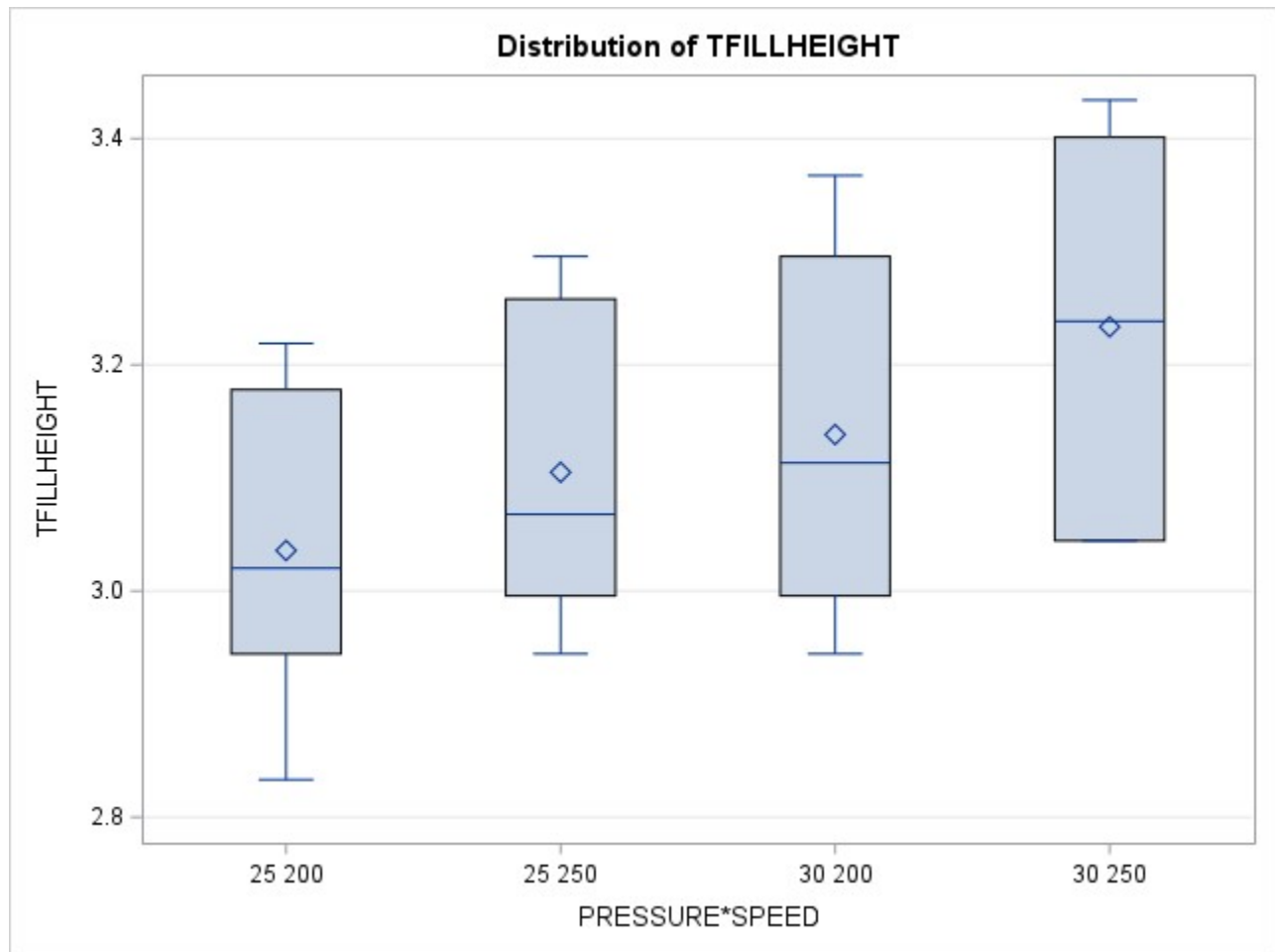
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.001457
Critical Value of Studentized Range	3.08118
Minimum Significant Difference	0.0339

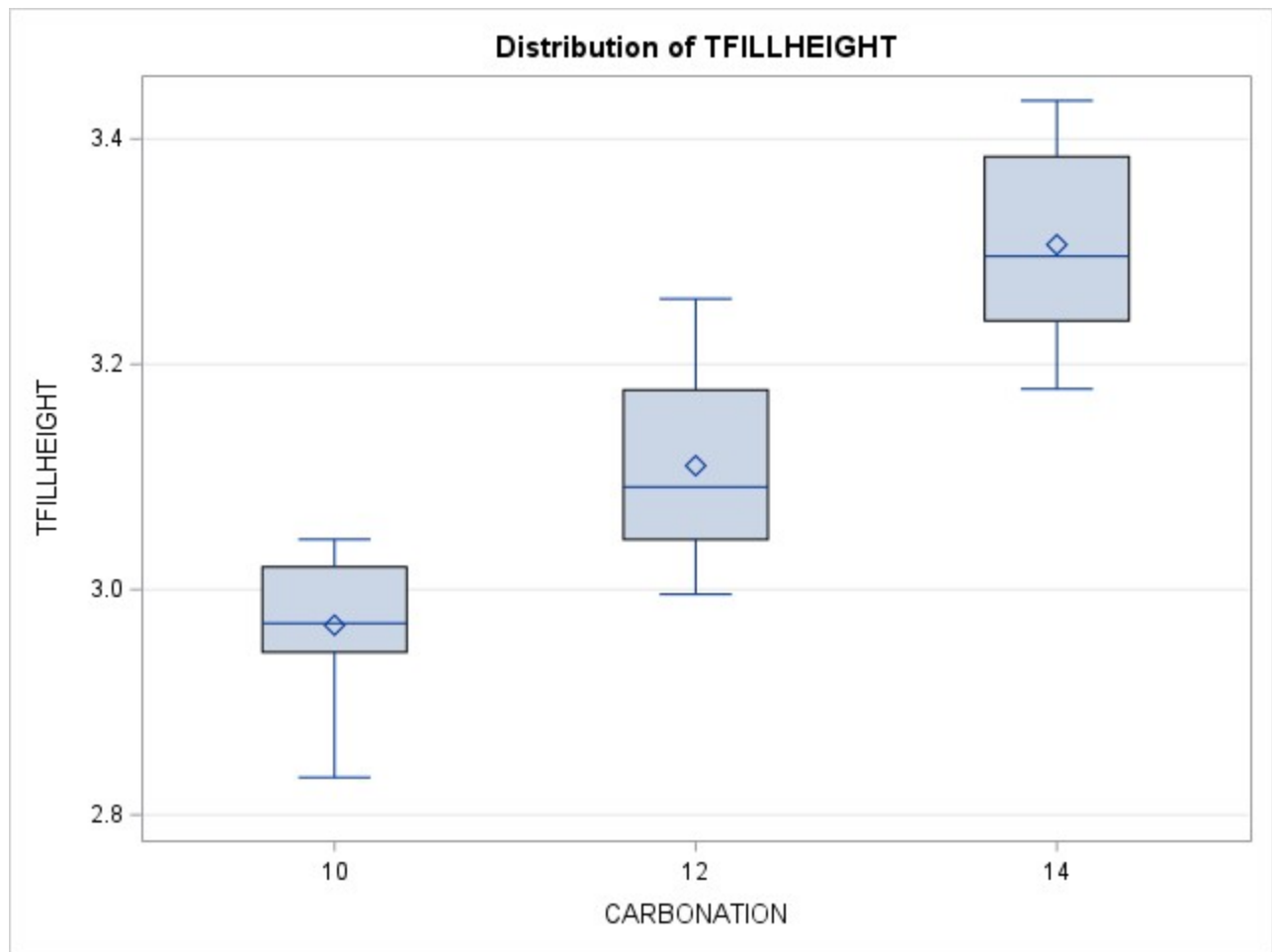


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The GLM Procedure



Level of PRESSURE	Level of SPEED	N	TFILLHEIGHT	
			Mean	Std Dev
25	200	6	3.03580611	0.14479130
25	250	6	3.10494492	0.14242813
30	200	6	3.13830677	0.16577440
30	250	6	3.23353364	0.16766718



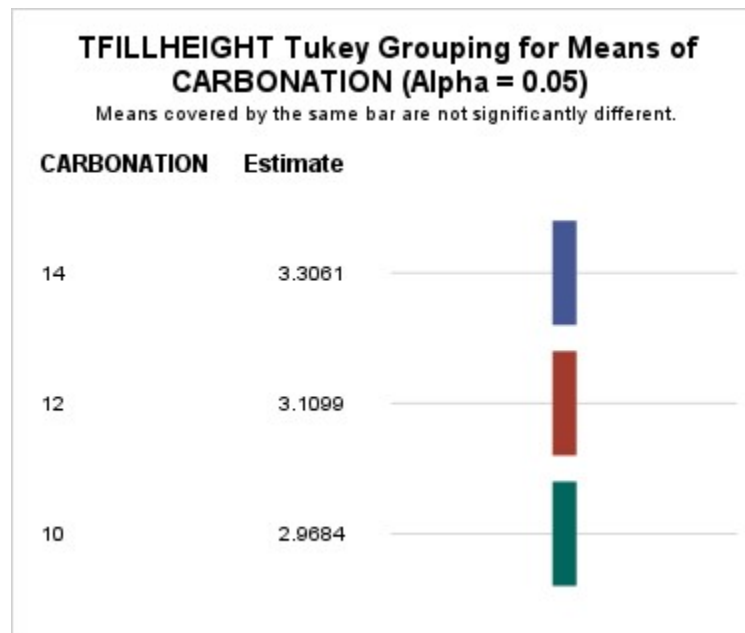
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The GLM Procedure

Tukey's Studentized Range (HSD) Test for TFILLHEIGHT

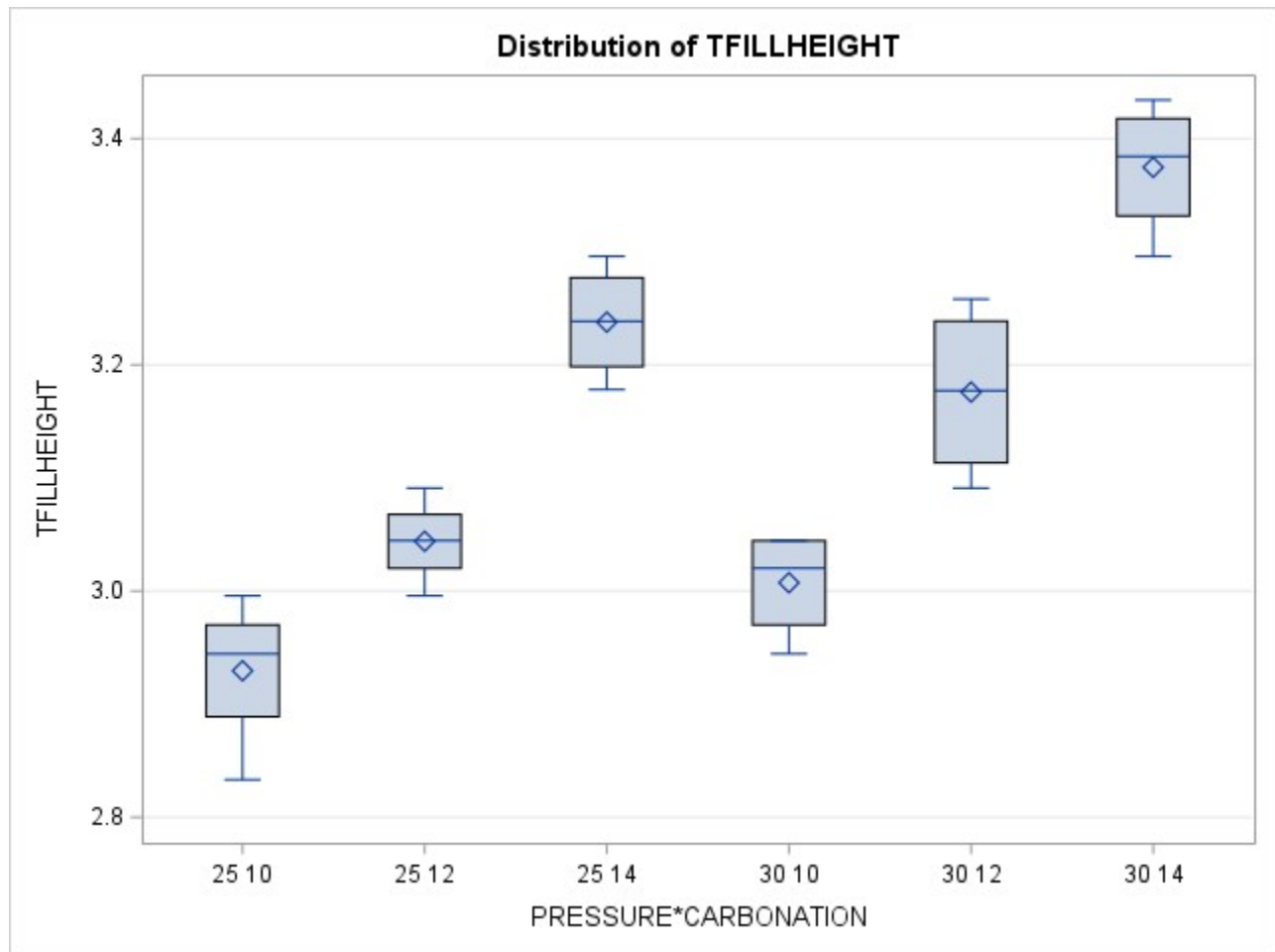
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.001457
Critical Value of Studentized Range	3.77289
Minimum Significant Difference	0.0509

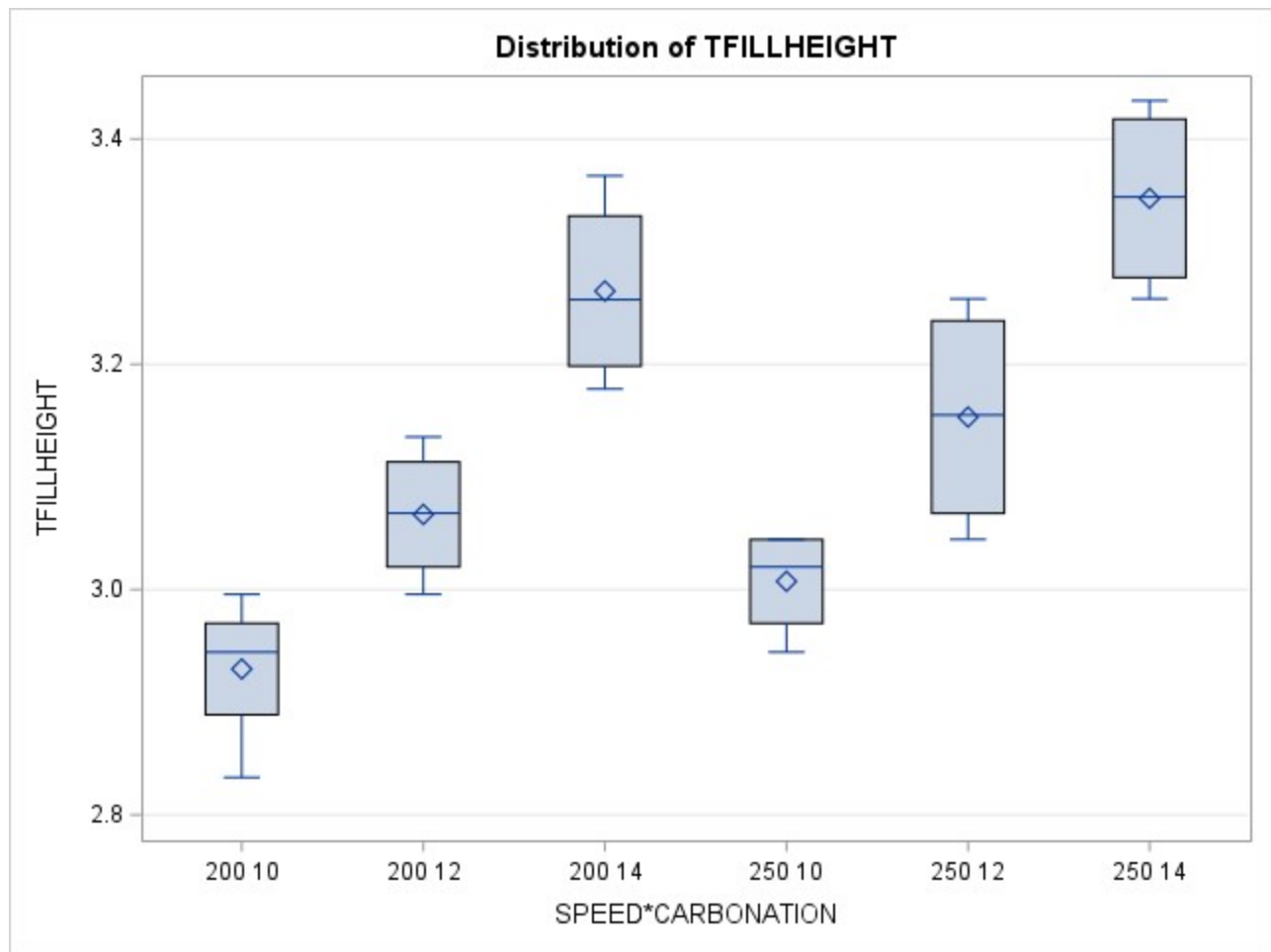


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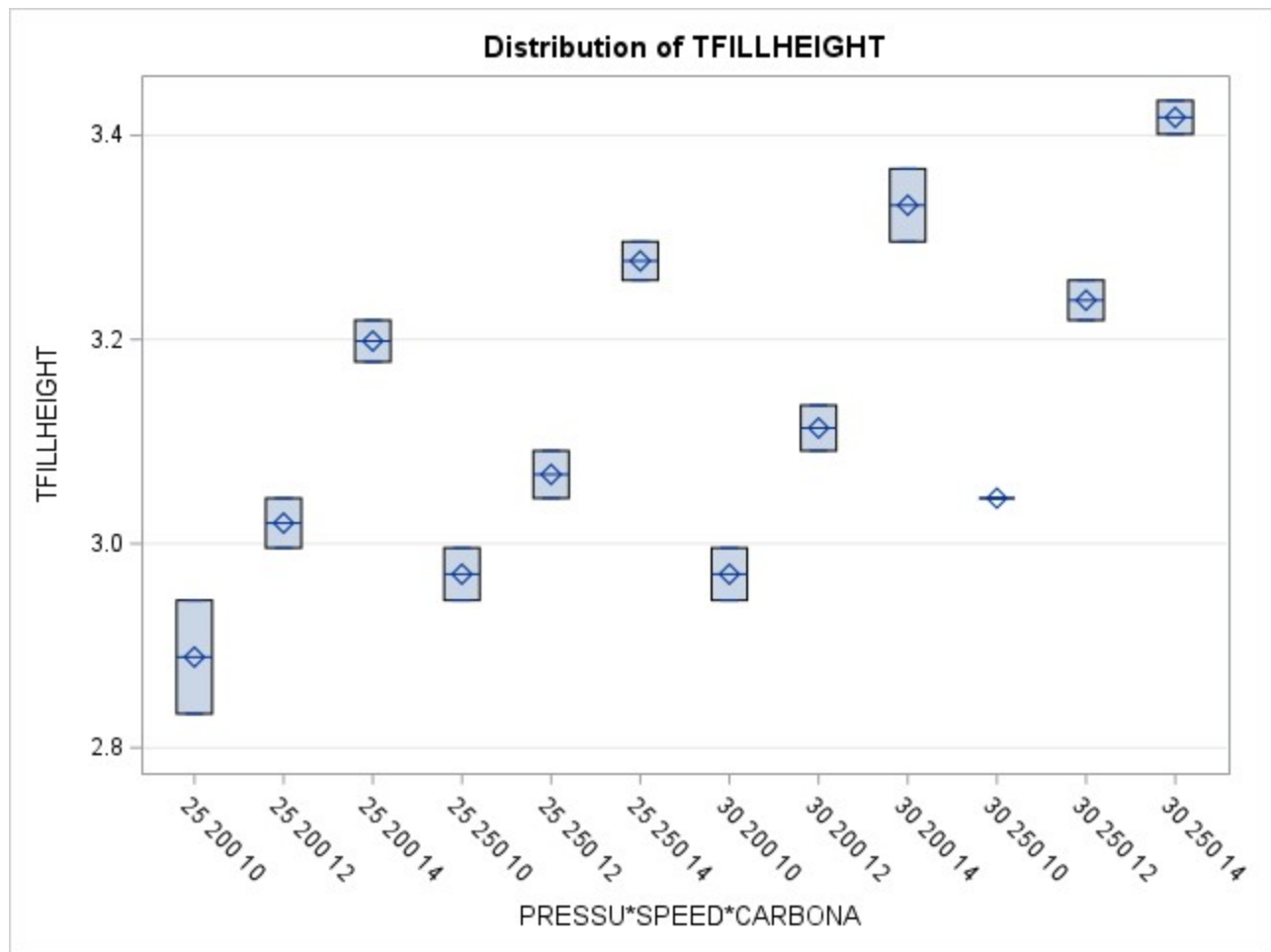
The GLM Procedure



Level of PRESSURE	Level of CARBONATION	N	TFILLHEIGHT	
			Mean	Std Dev
25	10	4	2.92945589	0.06856669
25	12	4	3.04395490	0.03891574
25	14	4	3.23771576	0.05068836
30	10	4	3.00730403	0.04780635
30	12	4	3.17587726	0.07623783
30	14	4	3.37457932	0.05913612



Level of SPEED	Level of CARBONATION	N	TFILLHEIGHT	
			Mean	Std Dev
200	10	4	2.92945589	0.06856669
200	12	4	3.06669785	0.06014832
200	14	4	3.26501559	0.08386979
250	10	4	3.00730403	0.04780635
250	12	4	3.15313431	0.10163817
250	14	4	3.34727950	0.08371640



Level of PRESSURE	Level of SPEED	Level of CARBONATION	N	TFILLHEIGHT	
				Mean	Std Dev
25	200	10	2	2.88882616	0.07864840
25	200	12	2	3.02012736	0.03449986
25	200	14	2	3.19846483	0.02886551
25	250	10	2	2.97008563	0.03626984
25	250	12	2	3.06778245	0.03289462
25	250	14	2	3.27696670	0.02668644
30	200	10	2	2.97008563	0.03626984
30	200	12	2	3.11326833	0.03143214
30	200	14	2	3.33156635	0.05052912
30	250	10	2	3.04452244	0.00000000
30	250	12	2	3.23848618	0.02773323
30	250	14	2	3.41759229	0.02318591

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The UNIVARIATE Procedure
Variable: RESID

Moments			
N	24	Sum Weights	24
Mean	0	Sum Observations	0
Std Deviation	0.02756979	Variance	0.00076009
Skewness	0	Kurtosis	-0.8017669
Uncorrected SS	0.01748214	Corrected SS	0.01748214
Coeff Variation	.	Std Error Mean	0.00562766

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	0.02757
Median	0.00000	Variance	0.0007601
Mode	-0.02565	Range	0.11123
		Interquartile Range	0.04549

Note: The mode displayed is the smallest of 3 modes with a count of 2.

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > t 	1.0000
Sign	M	0	Pr >= M 	1.0000
Signed Rank	S	0	Pr >= S 	1.0000

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.93348	Pr < W	0.1166
Kolmogorov-Smirnov	D	0.182301	Pr > D	0.0384
Cramer-von Mises	W-Sq	0.165217	Pr > W-Sq	0.0147
Anderson-Darling	A-Sq	0.880238	Pr > A-Sq	0.0215

Quantiles (Definition 5)	
Level	Quantile

100% Max	0.0556128
99%	0.0556128
95%	0.0357295
90%	0.0256466
75% Q3	0.0227429
50% Median	0.0000000
25% Q1	-0.0227429
10%	-0.0256466
5%	-0.0357295
1%	-0.0556128
0% Min	-0.0556128

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-0.0556128	1	0.0243951	4
-0.0357295	17	0.0256466	8
-0.0256466	13	0.0256466	14
-0.0256466	7	0.0357295	18
-0.0243951	3	0.0556128	2